Experiment - 7

Implement packet sniffing using Raw sockets in python

Aim:

To implement a packet sniffing using raw sockets in python capture and analyze network packets

Procedure:

1) open terminal with root previledge

2) create a python script

3) import socket, struct, textwrap and binascii modules

4) create a raw socket using socket AF_PACKET (linux) or sockets. AF_INET with socket

5) Bind the socket to the network interface

6) use a loop to continously recive packets with socket.recvfrom()

7) upack the ethernet header

command / code:

```
# packet_sniffer.py

import socket, struct, binascii

def mac_addr(raw):
    return ':'.join('%02x' % b for b in raw)

s = socket.socket(socket.AF_PACKET,
    socket.SOCK_RAW, socket.ntohs(0x0003))

while True:
    raw_addr = s.recvfrom(65535)
    eth = struct.unpack('!6s6sH', raw_addr[0][:14])
    dest = mac_addr(eth[0]); src = mac_addr
        (eth[1]); proto = eth[2]
    print('src: %s dst: %s
        proto: %s' % (src, dest,
        proto, '%04x'))
```

8) Extract and print the source mac address

9) for IP packets (Ether type 0x800)

10) Display source IP, destination IP, TTL, and the protocol number

Result:

The packet sniffer was successfully implemented. Ethernet frames and IP packet were captured and source/destination MAC and IP address along with protocol information were displayed